ESP32 firmware v2.2.0: the safety logic runs on core 1, all networking on core 0.
They share only two bounded queues and one mutex-guarded status snapshot.

SENSING (inputs)

PIR motion sensor
HC-SR501, OUT on GPIO 13
10 s warm-up, 150 ms confirm

Door / window switch
reed or slide switch, GPIO 14
internal pull-up; HIGH = open

ARM / DISARM button
momentary, GPIO 27
internal pull-up; LOW = pressed

ESP32 DevKit — dual core, FreeRTOS

Core 1: safety loop (Arduino loop task)
never blocks, watchdog-supervised, works offline

1. Input conditioning
debounce 40 ms on the door and the button
PIR: 10 s warm-up, 150 ms confirm

2. State machine
DISARMED → EXIT_DELAY 10 s → ARMED
ARMED + motion or door → ENTRY_DELAY 10 s
ENTRY_DELAY expires → ALARM 30 s → ARMED
button arms and disarms; remote: ARM, DISARM, STATUS
armed state kept in NVS across power cuts

3. Output driver
buzzer tone 2.7 kHz, two LEDs,
OLED refresh every 250 ms (4 Hz)

Command
queue, 4 deep

Event
queue, 16 deep
drop oldest

Status
snapshot 500 ms
safety: 0 ms
network: 10 ms

Core 0: network task
slow DNS or TCP can never stall the alarm

Wi-Fi manager
retry every 15 s
NTP clock (IST)

MQTT client
Last Will “offline”
reconnect every 5 s
subscribes to …/cmd
Home Assistant discovery

Event publisher
JSON events, sequence no.
oldest first
retained state,
heartbeat every 30 s

Wi-Fi, MQTT 1883 / 8883 TLS

LOCAL OUTPUTS

Buzzer
passive piezo
GPIO 23
2.7 kHz

Status LEDs
green GPIO 26
red GPIO 25

OLED SSD1306
I2C 0x3C
GPIO 21 / 22

NETWORK AND USER SIDE

MQTT broker
default: public test broker
evaluated against a private broker
with TLS, per-client ACLs, credentials
topics: availability, state, event, cmd

state, events ↓ ↑ commands

Homeowner clients
Home Assistant / Node-RED / dashboard
phone MQTT app: alert on an alarm event

LEGEND

- Sensing
- Safety core (core 1)
- Network core (core 0)
- Shared queues
- Local outputs
- Network and user side

Events raised while offline wait in the queue (up to 16) and are sent on reconnect. A remote DISARM needs a valid HMAC tag over a fresh, NVS-persisted counter.